



SAPIENZA
UNIVERSITÀ DI ROMA

Visual Analysis for Formula 1 Race Strategies

Visual Analytics Project

A.Y. 2025/2026

Angela Di Giampaolo
2205621

Bogdan Andrei Tutuianu
2217841





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Introduction



Formula 1 is a motorsport sector characterized by the management and analysis of a huge amount of data (car telemetry, tyre condition, lap times, track temperature).



These data needs to be analyzed in order to fully understand the dynamics characterizing a specific Formula 1 Race.



Our goal is to develop a dashboard that can be used by a **race engineer** responsible for evaluating post-race data and comparing different strategies adopted by the teams.

The tool can be also used by **sport journalists** that can craft data-driven technical narratives.



Related Works

Author	Content	Our proposal
Lampprecht et al.	Interactive web-based tool providing a yearly calendar overview and detailed lap-time comparison	It guarantees a deeper tactical dive into a single race, integrating also a Strategy Gantt and a PCA view
Eckl et al.	Framework for analyzing F1 seasons with synchronized views, allowing for filtering by driver and season	Differently from Eckl's system, that is primarily descriptive, our proposal is white box and exploratory since it integrates real-time linear regression for tyre degradation triggered by user selection



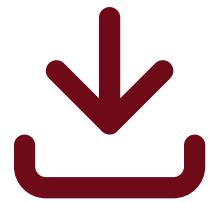
Related Works

Author	Content	Our proposal
Msakamali	Investigation of the complexities of analyzing team-specific tactical implementations through telemetry and tyre performance case studies.	Differently from Msakamali's static analysis, our solution offers a highly coordinated environment, that allows for the analysis of technical elements that are treated separately in Msakamali's work
Mugge and Zandieh	Exploration of importance of tyre management, performed using FastF1 and aiming to identify the optimal window lap life	While Mugge and Zandieh focuses on the predictive aspect, our tool focuses on the interpretative side.

Dataset and Domain



The data we used comes from **Fast F1**, an open-source Python package that provides access to Formula 1 timing data and car telemetry.



We downloaded data related to **Monza 2024 Grand Prix**.

The dataset provides a lap-by-lap breakdown for each driver participating to the race. It consists of approximately 1,000 rows (20 drivers by 53 laps) and 29 columns, resulting in an **AS Index of 29232**.

The data dimensions are 4:

- Driver Information
- Performance Metrics
- Lap Context
- Track and Weather Conditions

Preprocessing



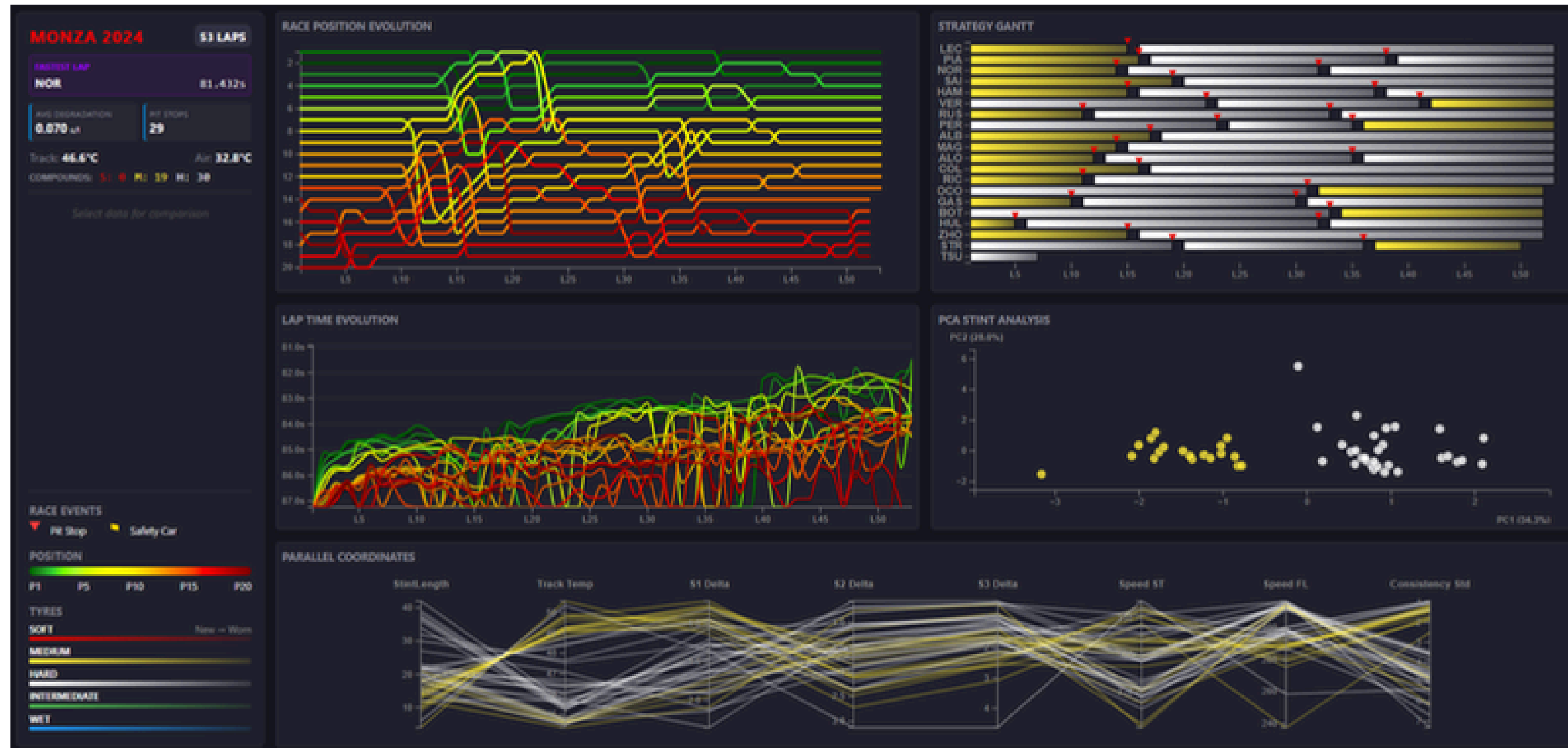
The system groups individual laps into **stints**, consecutive lap on the same tyre set. For each stint, a set of aggregated features is computed to summarize the driver's performance.

In addition, several derived metrics are calculated:

- **Tyre Degradation:** Estimated using linear regression on lap times within a stint. The slope of the regression line represents the pace drop-off per lap.
- **Performance Deltas:** Average sector deltas are computed to compare driver efficiency across different sections of the track.
- **Consistency:** The standard deviation of lap times is calculated to measure the driver's consistency during the stint.

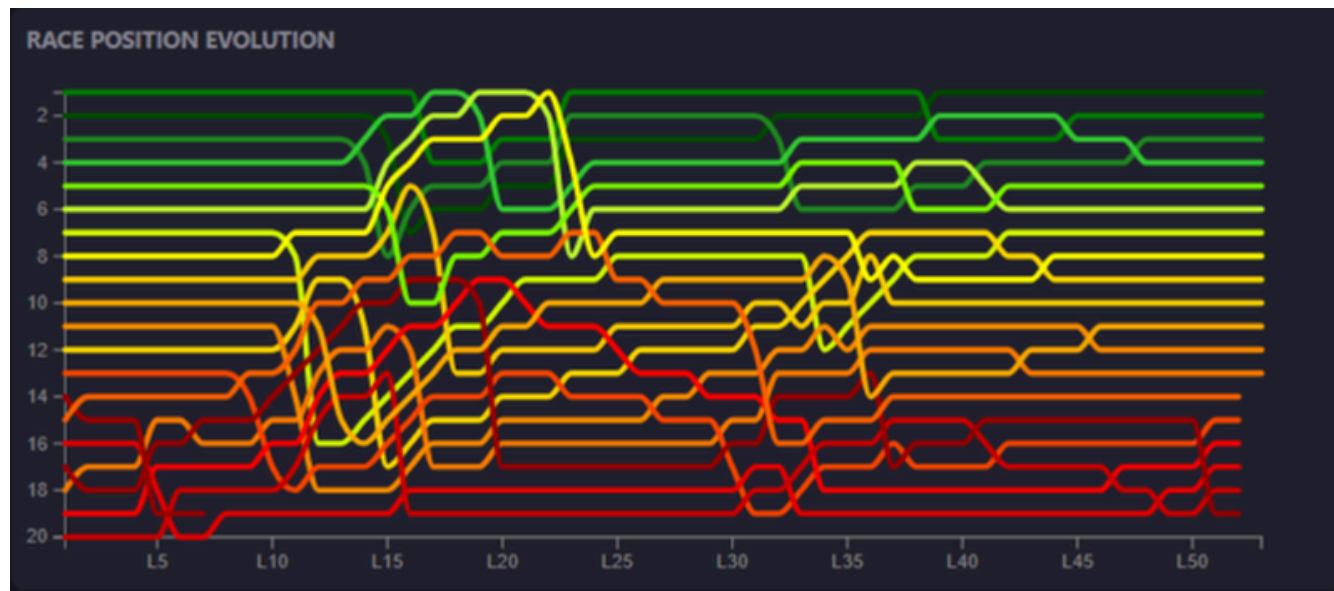


Application Overview



Application Overview

Race Position Evolution:



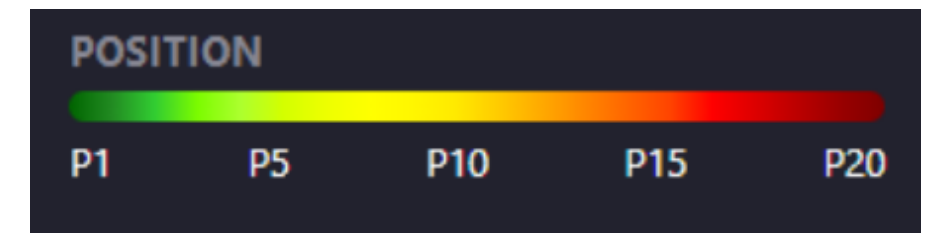
Linechart representing the evolution of the driver's track position throughout the race

It adopts the following encoding:

- x-axis → lap number
- y-axis → race position

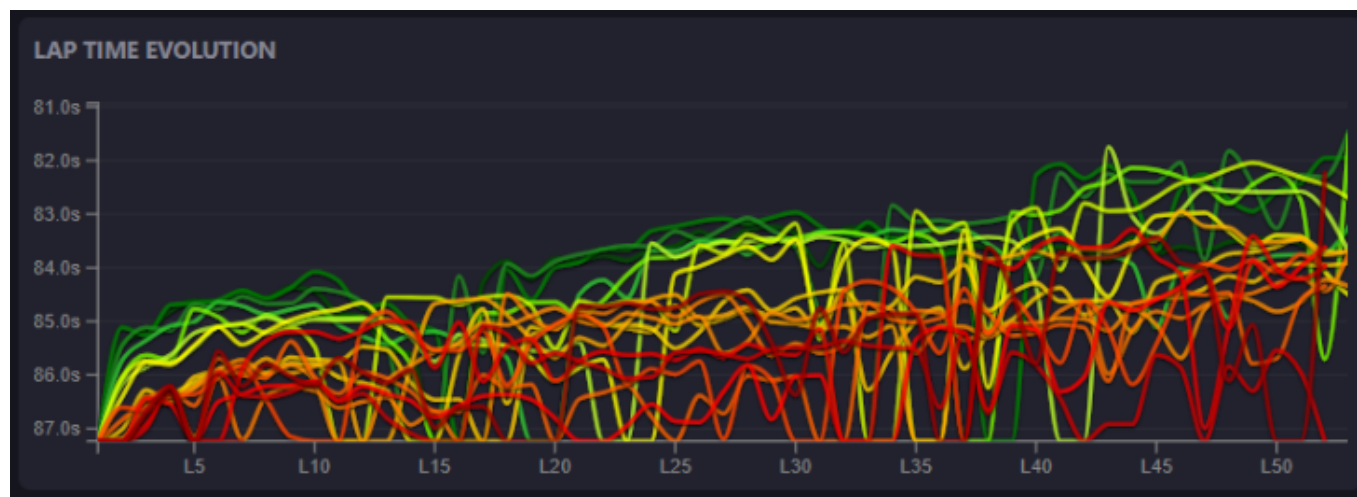
Drivers are identified through a continuous color scale where the two extremes are:

- red → for the driver in last position
- green → for the race winner



Application Overview

Lap Time Evolution:



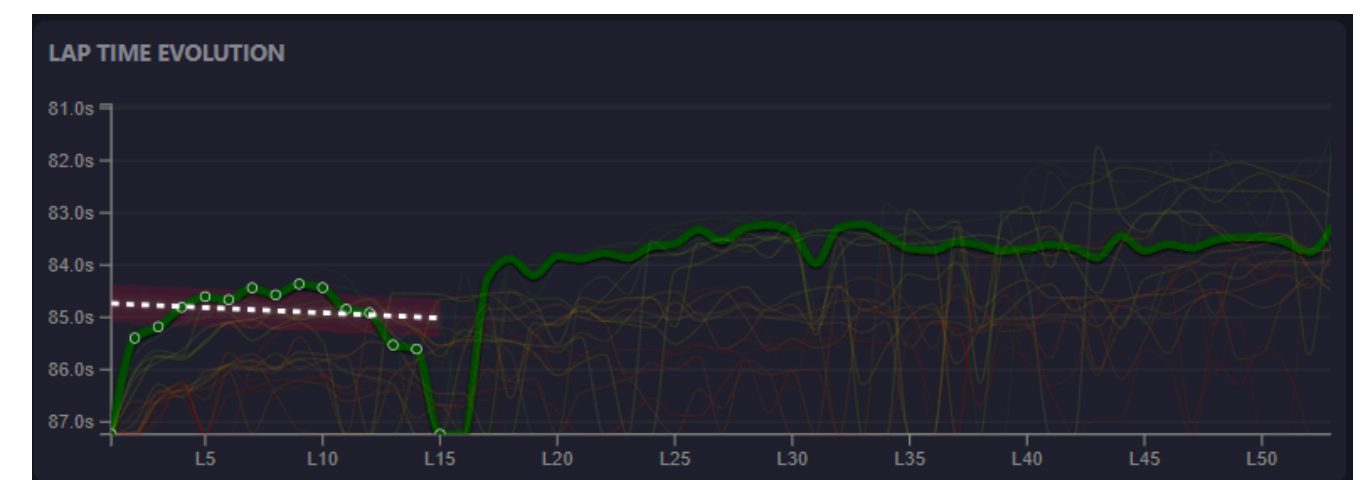
Linechart representing the pace of the drivers.

It adopts the following encoding:

- x-axis → lap number
- y-axis → registered lap time

Drivers are encoded as done in Race Position Evolution

A visualization of the linear regression of lap times within a stint allows to have a visual quantification of pace drop-off over time



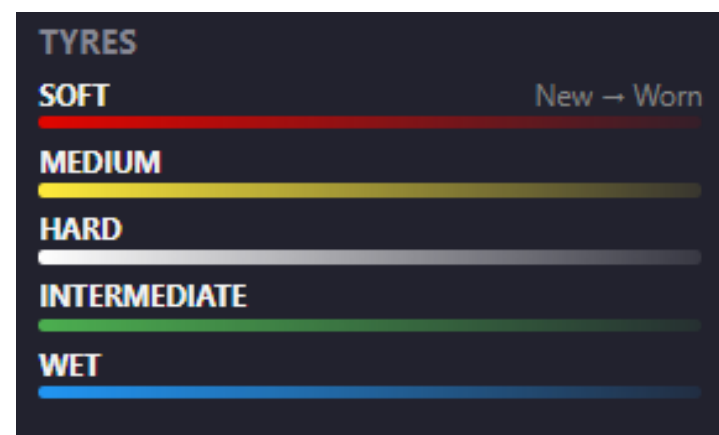
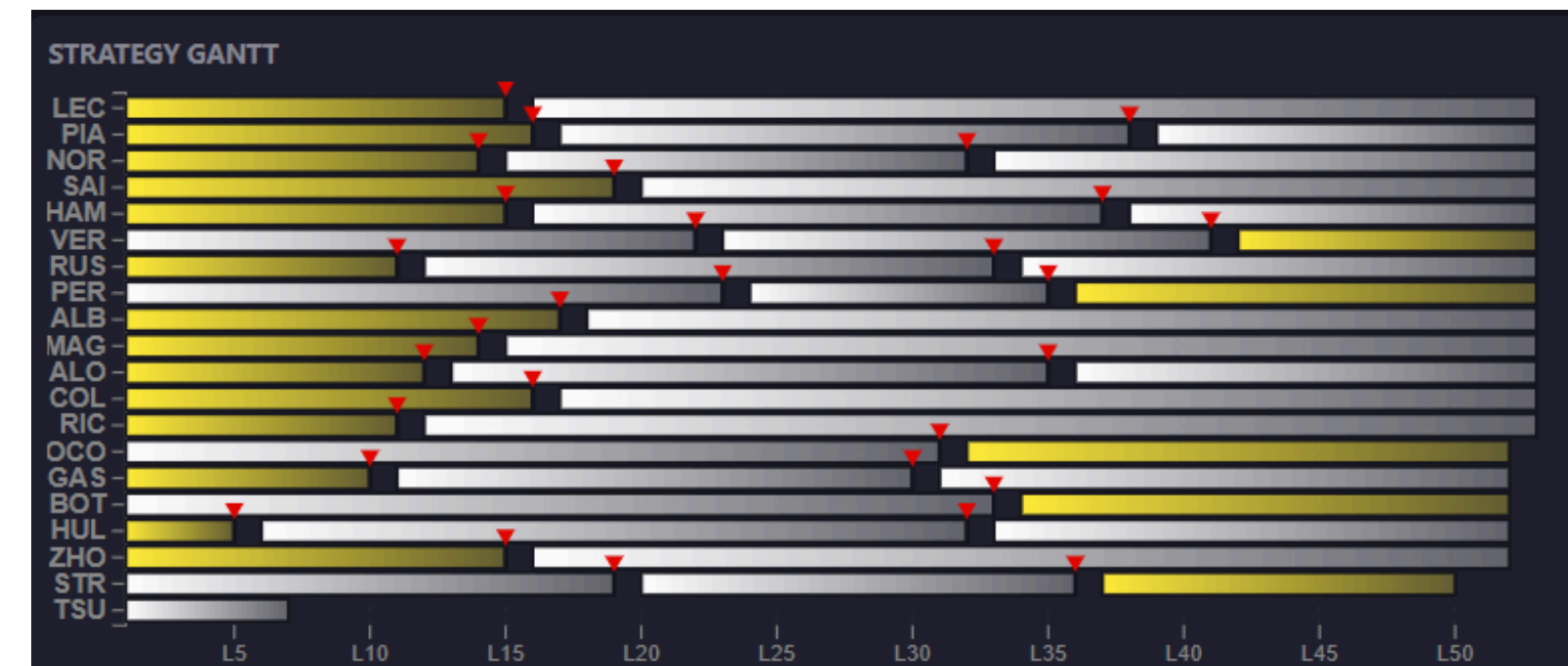
Application Overview

Strategy Gantt:

It allows to illustrate the race strategies adopted by the grid.

The encoding is the following:

- x-axis → laps
- y-axis → drivers ordered by final race result
- horizontal bar → stint
- red reversed triangle → pit stop



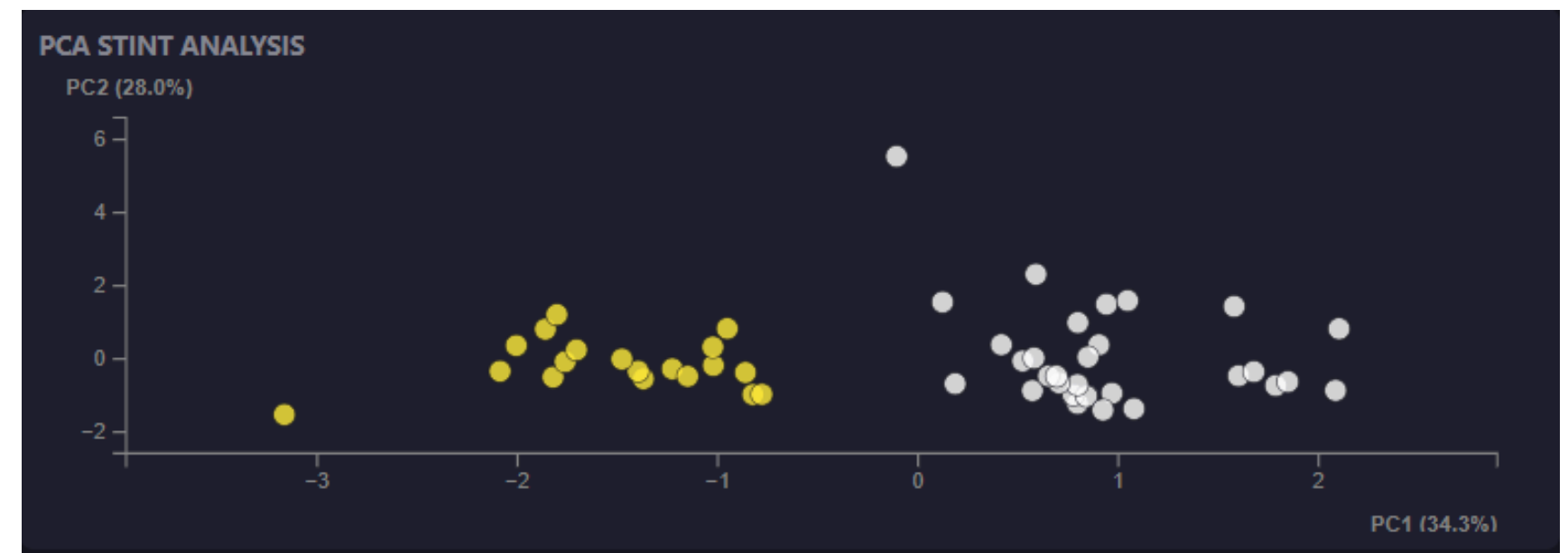
The visual variables for the bars follows the Formula 1 tyre domain convention, while the luminance of each bar encodes the progressive consumption of the tyre.

Application Overview

PCA Stint Analysis:

It is a scatterplot visualizing the results of PCA applied to stint features.

Each point represents a single stint and it is positioned according to the first two principal components.



Points are color-coded based on the tyre compound used.

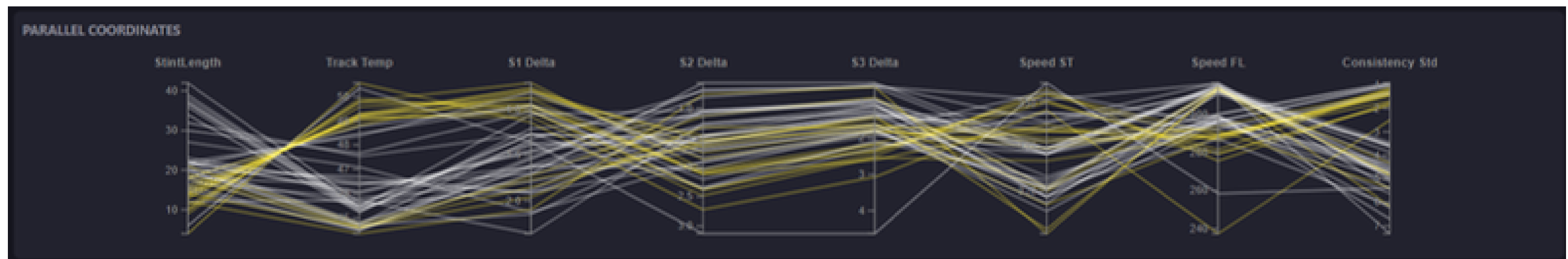


Thanks to this representation the user can identify clusters or it can observe outliers.

Application Overview

Parallel Coordinates:

It allows to inspect the multivariate attributes of the stint



The encoding is the following:

- each polyline represents a single stint and it is color-coded according to the type of tyre used in that stint
- each vertical line represents a quantitative metric (Stint Length, Track Temperature, Sector Deltas, Speed Traps, Consistency Std).



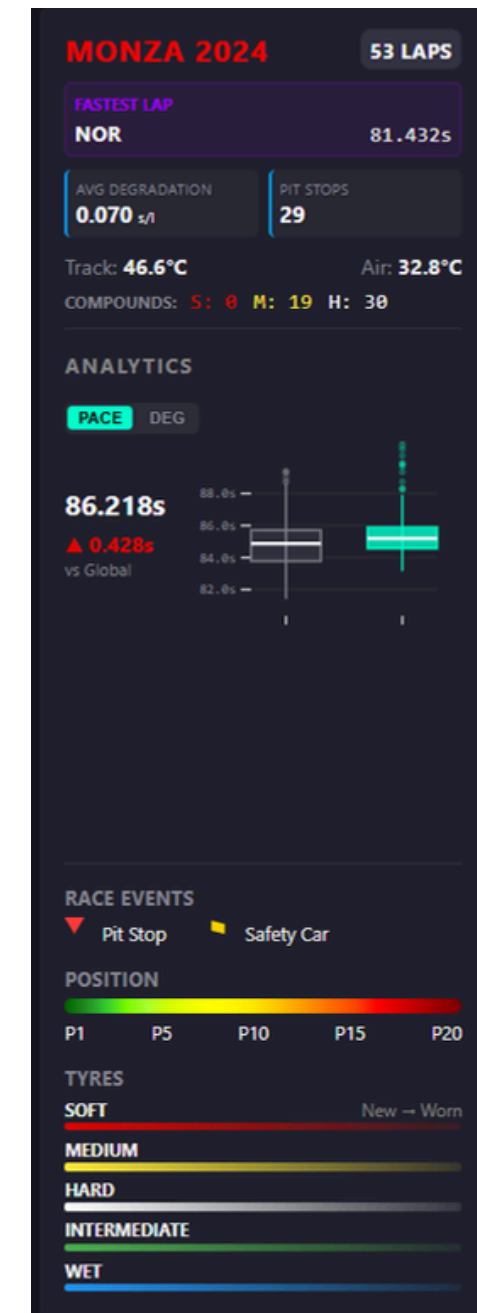
Application Overview

Sidebar:

This component acts as control and summary panel.

It is divided into three sections:

- General Statistics → it contains aggregate context about the race
- Analytics (Details on Demand) → this section automatically generates comparative box plots based on user selections
- Legend → it describes the visual encoding adopted throughout the dashboard.



The One-Stop Strategy Efficiency

The race winner, Charles Leclerc, adopted a one pit stop strategy



Oscar Piastri adopted a two pit stop strategy and arrived second despite starting from pole position



Lap 1-15: Piastri is faster than Leclerc → Lap 15: Leclerc's pit stop → Lap 16: Piastri's pit stop

Lap 38: Piastri's pit stop due to augmenting lap times

← Lap 16-38: lap times of two drivers converge



After the pit stop Piastri was faster than Leclerc, but he lost time during the pit stop and he was not able to reach the race winner before the end of the race

PCA Outlier Analysis

The PCA Stint Analysis allows to identify two technical anomalies:

- **Hulkenberg's Initial Stint** → in the PCA scatterplot most Medium tyre stint are clustered together, but this does not hold for Nico Hulkenberg's first stint.



This point is isolated due to a minimum stint length (only 5 laps) and inferior performances in S2 and S3, combined with the worst registered Speed FL value

- **Tsunoda's Technical Failure** → the dashboard clarified this anomaly because it is the only hard stint lasting only 7 laps. Tsunoda also registered the worst Sector 2 and 3 performances, despite decent Speed ST value, and this highlights his technical failure



Importance of Driver's Ability

Zhou and Ricciardo are isolated from all the other drivers who performed a one-pit stop strategy (Medium-Hard) and that used the hard compound for a high number of laps



This is due to both drivers failing to reach the same speeds (SpeedST) as the leaders within the cluster

In addition, Zhou, compared to top-tier drivers, shows an inability to manage the hard compound.



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**THANK YOU
FOR
THE ATTENTION!**